DRUGS · LOCAL ANAESTHETICS AND VASOCONSTRICTORS

Cocaine (topical)

Cocaine is the only local anaesthetic that also causes intense vasoconstriction, by blocking noradrenaline reuptake. That made it the classic preparation for nasal surgery: Moffett's solution or cocaine paste shrinks and anaesthetises the nasal mucosa in one step. It is also a sympathomimetic with real cardiovascular risks, a controlled drug, and a cause of septal perforation and midline destruction when abused. Many units now use lidocaine with adrenaline or oxymetazoline instead.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

COCAINE (TOPICAL)

	SUMMARY
Class	Ester local anaesthetic and sympathomimetic
How it works	Sodium-channel block plus inhibition of noradrenaline, dopamine and serotonin reuptake, causing vasoconstriction and central stimulation
Preparations	Topical solution 4% to 10%; paste 10% to 25%; Moffett's solution (classically 2 ml of 8% cocaine, 2 ml of 1% sodium bicarbonate and 1 ml of 1:1,000 adrenaline, applied on pledgets or as a spray, with the patient in the head-down position)
ENT uses	Topical anaesthesia and decongestion for septal, turbinate, sinus and nasal fracture surgery
Maximum dose	1.5 mg/kg (many units limit the total to 100 mg in an adult)
Avoid	Ischaemic heart disease, hypertension, arrhythmia, hyperthyroidism, phaeochromocytoma, pregnancy; patients on MAOIs, tricyclics or non-selective beta blockers
Legal status	Schedule 2 controlled drug: record, store and dispose of it as required

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Application

Apply on pledgets or as a measured spray, keep within the dose limit, and allow 10 to 15 minutes. With Moffett's solution, adrenaline adds vasoconstriction and the bicarbonate improves absorption of the cocaine base.

2.2 Recognising misuse

Chronic intranasal cocaine use causes septal perforation, loss of the turbinates, palatal perforation and destruction of the midline (cocaine-induced midline destructive lesion), which can mimic granulomatosis with polyangiitis, including a positive ANCA (often PR3 or HNE). Ask about drug use, and be aware that levamisole adulteration causes vasculitis and neutropenia.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Tachycardia, hypertension, arrhythmia, coronary vasospasm, myocardial infarction, stroke, agitation, hyperthermia and seizures; with adrenaline, the risk is additive. Treat toxicity with benzodiazepines, cooling and avoidance of beta blockers (unopposed alpha stimulation).

3.2 Interactions

MAOIs, tricyclics, adrenaline and other sympathomimetics; non-selective beta

blockers; halothane (arrhythmias).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism of vasoconstriction (reuptake inhibition).
- Moffett's solution and its components.
- The maximum dose and cardiovascular contraindications.
- **Cocaine-induced midline destructive lesion** versus GPA.

4.2 Common pitfalls

COMMON PITFALLS

- Giving a beta blocker to treat cocaine-induced hypertension.
- Forgetting controlled drug procedures.

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